Mamoon Khan

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github.com/Moon-Khan Portfolio

Professional Summary

Full-Stack Software Engineer with 2 years of experience shipping production web applications end-to-end. Strong hands-on skills in **React, JavaScript, TypeScript, Node.js, and Express**, with a solid backend foundation in **REST API design, PostgreSQL/MongoDB data modeling, multi-tenant architecture, and application security**. Experienced in **JWT authentication, RBAC, query optimization, automated testing, and CI/CD**, with additional backend depth in ASP.NET Core, C#, and Entity Framework Core.

Technical Skills

- **Frontend:** React, JavaScript, TypeScript, HTML, CSS, Responsive Web Applications, Next.js, Component-Based UI
- **Backend:** Node.js, Express.js, NestJS, REST APIs, GraphQL, Web APIs, Microservices, Background Jobs, Async Programming
- **Databases:** MongoDB, Mongoose, PostgreSQL, MySQL, Microsoft SQL Server, Redis, Row-Level Security, Query Optimization, Indexing, Connection Pooling
- **Microsoft / .NET:** C#, ASP.NET Core, ASP.NET Core Web API, Entity Framework Core, LINQ, Clean Architecture, CQRS
- **Security:** JWT Authentication, Refresh-Token Rotation, Role-Based Access Control (RBAC), OAuth, SSO, Secure API Design, Application Security
- **DevOps / Tools:** Git, GitHub, GitHub Actions, Docker, Vercel, AWS EC2, CI/CD Fundamentals, Postman, Visual Studio

Experience

CodingCops Lahore, Pakistan
Software Engineer *Dec 2024 – Present*

- Own web application features end-to-end, from idea to production, building **React/JavaScript** frontends and **Node.js** and **ASP.NET Core** backends for scalable, business-critical workflows.
- Designed and shipped RESTful APIs and backend services, integrating third-party platforms and handling authentication, subscriptions, data synchronization, and end-to-end business flows.
- Optimized SQL Server and PostgreSQL queries, indexes, joins, and connection pooling to improve API performance and keep applications responsive under concurrent load.

Agrilift Lahore, Pakistan
Support Engineer *Aug 2024 – Dec 2024*

- Diagnosed, debugged, and resolved production issues across web applications, improving application stability and reducing service disruptions.
- Optimized database queries, stored procedures, and data models for core business modules to improve reliability and performance.
- Investigated application and database issues across the software lifecycle, identifying root causes and implementing maintainable fixes.

TenX Lahore, Pakistan
Software Developer Intern *Jun 2024 – Aug 2024*

- Developed web application modules and internal business workflows, implementing UI, CRUD functionality, validation, and business logic.
- Built and integrated RESTful Web APIs and third-party services into application features.
- Debugged application issues and contributed to maintaining and improving existing software modules.

Selected Projects

BBB Smart System *Node.js / NestJS + GraphQL Federation*

- Architected and built the Dispatcher microservice from scratch using **NestJS** and **GraphQL Federation**, enabling field teams to create, assign, search, and report on incidents across a distributed schema.

- Load-tested the service at 19,000+ requests across 15 endpoints with 35 concurrent users while holding a 0.02% error rate, and diagnosed a reporting endpoint driving database CPU utilization to 98%.
- Optimized query performance under concurrency with targeted indexes on high-filter fields, rewriting the core listing query with joins in place of N+1 patterns, and implementing connection pooling with query and statement timeouts.

Multi-Tenant Management Backend *NestJS + PostgreSQL (RLS) + Redis + Docker*

- Built a self-hosted multi-tenant SaaS backend with **NestJS**, **PostgreSQL 16**, and **Redis**, using a clean/hexagonal layout (domain, use-cases, infrastructure) and a unified response envelope with centralized exception handling.
- Enforced tenant isolation at the database level with **PostgreSQL Row-Level Security** and a non-owner runtime DB role, resolving tenant context from the verified JWT via AsyncLocalStorage and short per-request transactions that release connections before hashing, signing, or responding.
- Implemented **JWT** authentication with refresh-token rotation and reuse detection (a replayed token revokes its whole family) and **RBAC** (admin/member/viewer) through a global role guard, blocking self-assigned roles at registration.
- Added per-tenant **Redis** rate limiting using atomic Lua scripts, with per-IP overrides on login and register and standard `X-RateLimit` headers.
- Wrote 32 Jest unit tests and 23 Supertest e2e tests against real Postgres/Redis containers, verifying tenant isolation, RBAC, token rotation, and rate limiting, with a **GitHub Actions** CI pipeline running lint, build, and tests.

MyValueApp Tracking Platform *React + Node.js + ASP.NET Core 8*

- Delivered full-stack features across **React** frontends and **Node.js** and **.NET** backend services, owning 24 API modules end-to-end spanning recognitions, skills, 360° assessments, development plans, dashboards, and shareable profiles.
- Designed and shipped subscription billing and SSO integration with GoHighLevel from the ground up, including new endpoints, contact and subscription lifecycle handling, and schema migrations that introduced recurring billing and login flows the product previously lacked.
- Resolved 98+ production bugs across authentication, dashboards, and data-sharing modules, and added 12 automated test suites to keep a ∼35K-line codebase stable through weekly module releases.
- Implemented JWT-based authentication and role-based access control, and improved API performance through query optimization across both Node.js and .NET services.

Chat Flow — Real-Time Chat App *Node.js/Express + MongoDB + Socket.IO*

- Built the backend for Chat Flow, a real-time messaging app, using **Node.js/Express** and **MongoDB/Mongoose**, with **JWT** authentication and email verification (MailerSend) for secure sign-up and login.
- Implemented live messaging with **Socket.IO** — room joins, message send/delete events, and online-presence updates broadcast across connected clients.
- Designed REST APIs for users, conversations, and stories, supporting group and private chats, conversation pinning, and multi-file uploads (up to 10 files) via Multer.
- Deployed as a live full-stack web app on **Vercel** (live demo) with environment-based CORS and production email delivery.

Education

FAST – National University of Computer and Emerging Sciences Pakistan
Bachelor's in Computer Science

Certifications

- AWS Cloud Practitioner Essentials – AWS Training and Certification
- Microsoft Learn AI Skills Challenge – Microsoft
- Supervised Machine Learning: Regression and Classification – DeepLearning.AI, Coursera, Stanford